

Amir Elrefai Mostafa Elfalw

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Professional Profile

AI & Computer Vision Engineer with strong expertise in medical imaging, multi-modal feature fusion, and RAG-based systems. Skilled in building end-to-end deep learning pipelines, integrating AI with embedded hardware, and deploying intelligent AI solutions. Passionate about solving real-world problems with AI and IoT technologies.

Technical Skills

- **AI & Deep Learning:** Medical Image Analysis, Image Segmentation (U-Net, Attention U-Net), Object Detection (YOLOv8/v11), Multi-modal Feature Fusion, NLP, Transformers, RAG (LangChain + Pinecone).
- **Frameworks & Libraries:** Python, TensorFlow, Keras, PyTorch, OpenCV, MediaPipe, Scikit-learn, Pandas, NumPy, Matplotlib, Hugging Face Transformers.
- **Hardware & Embedded Systems:** Arduino, Raspberry Pi, ESP32, IoT sensors, Real-time gesture recognition, Edge AI deployment.
- **Deployment & MLOps:** Flask/FastAPI, Docker, Render, Model optimization (ONNX, TensorRT, pruning/quantization).
- **Programming & Data:** Python (Expert), C/C++, SQL, Data preprocessing, Feature engineering, Multi-modal datasets.

Projects

- **ThyraX - AI Medical Research System (Graduation Project)**
Developed a multi-stage pipeline for thyroid ultrasound analysis. Implemented **UNet++** (Nested UNet) for high-precision nodule segmentation. Integrated a **Medical RAG Assistant** using LangChain and Pinecone to provide clinical insights from patient metadata.
- **Driver Behavior Classification System (End-to-End)**
Built a real-time safety system using **ResNet50** to classify driver states (Safe, Distracted, Drowsy, Drinking) with 99.90% confidence. Developed the backend using **FastAPI** and deployed the API via **ngrok** for remote testing.
- **Real-time Traffic Monitoring & Analysis**
Engineered a high-speed vehicle detection and tracking system using **YOLO**. Optimized for high FPS to handle real-time urban traffic density and classification.
- **COVID-19 Diagnostic Tool (Transfer Learning)**
Implemented a **VGG16**-based model for automated detection from chest X-rays. Applied advanced **Data Augmentation** to improve generalization and overcome limited medical data constraints.
- **Sentiment Analysis Web App (NLP)**
Created an end-to-end NLP pipeline using **Multinomial Naive Bayes** for text classification. Built an interactive user interface using **Gradio** for real-time sentiment prediction (Positive/Negative).
- **Gesture-Controlled Smart Locker**
Integrated Computer Vision with **Arduino** to create a touchless biometric lock. Used MediaPipe for real-time hand gesture recognition to trigger physical actuators.

Courses & Certifications

- AI Track - NTI / Huawei Egyptian Talent Academy (80 hrs)
- CCNA - Digital Egypt Youth Program (120 hrs)
- Computer Vision Specialized Training - NTI (Ongoing, 72 hrs)
- Data Science Professional Track - DEPI (Ongoing, 6 months)
- IoT & Embedded Systems Internship - NTI (120 hrs)
- NVIDIA Deep Learning Institute (DLI) Training

Education

Faculty of Computer & Information Sciences, Mansoura University
B.Sc. in Computer Science

2022 – 2026 (Expected)
Current Grade: Very Good

Languages

- Arabic: Native | English: Advanced (Technical Proficiency)